MINI PROJECT PRESENTATION | IIIT DHARWAD

Medical Insurance Claim Fraud Detection Multi-Paradigm ML, Deep Learning, Agentic XAI & Browser Lab

Health Insurance Fraud Intelligence Platform: Leakage-Governed Benchmarks, 10-Model Evaluation & Dual-Path Deployment

Submitted by: B. Varshith (23BDS011) | J. Ganesh Kumar Reddy (23BDS024) | M. Jagadeeswar (23BDS033)

Supervised by: Prof. Ramesh Athe | Dept. of Computer Science & Engineering, IIIT Dharwad | November 2026

Project Scope & Unified Team Ownership

Integrating Classical ML, PyTorch Deep Learning, Agentic AI + XAI, Full-Stack App & Verified Research Lab

Approach 1 & Lab — J. Ganesh Kumar Reddy (Roll No: 23BDS024)

- Audited source Excel workbook (4,500 claims x 19 cols, SHA-256 verified).
- Identified upstream Cluster proxy leakage & ClaimAmount shortcut risk.
- Built 5 Classical ML baselines (LR, RF, XGBoost, SVM, GradientBoosting).
- Engineered submission/ 32-run 4-experiment ablation suite + Temporal rollout.
- Compiled pure-JS browser Claim Fraud Lab verified within 4.42e-7 parity.

Approach 2 & Full-Stack — M. Jagadeeswar (Roll No: 23BDS033)

- Designed & trained 5 PyTorch models: FraudMLP, ResidualMLP, CNN1D, TabTransformer & Autoencoder.
- Built unified FastAPI Inference Server (ml-service :8000) serving all 10 ML & DL models.
- Implemented Express proxy routes (/api/ml/*) & offline React fallback.
- Validated TabTransformer (99.11% PR-AUC) and documented Autoencoder negative study.

Approach 3 Agentic & Chat — B. Varshith (Roll No: 23BDS011)

- Built 6-Agent React pipeline UI & interactive Agentic Dashboard (:5173).
- Implemented Express /agentic/predict emitting 4-stage JSON verification trace.
- Integrated live Google Gemini 2.0 Flash Co-Pilot (POST /api/chat).
- Built Approach-3-Agentic-AI (:8020) XAIExplainer & SHA-256 hash scaffold.

Problem Motivation & Core Technical Challenges

Why Naive Binary Classification Fails on Real-World Healthcare Claims

1. Extreme Class Imbalance (6.00% Positives)

- 4,230 Legitimate (94.0%) vs. only 270 Fraudulent (6.0%) claims.
- An all-negative dummy classifier gets 94% accuracy while catching zero fraud.
- Requires weighted BCE (w+ = 15.67) and PR-AUC instead of accuracy alone.

2. Upstream Proxy Leakage & Shortcut Traps

- Post-hoc 'Cluster' isolates 267/270 frauds into Cluster 1 (inflated offline accuracy!).
- High-cardinality IDs (ClaimID, PatientID, ProviderID) risk overfitting.
- All frauds satisfy ClaimAmount >= $7,465.07—demanding explicit ablation stress tests.

3. Chronological Drift & Threshold Calibration

- Random splits leak future claim patterns backward into training folds.
- Fixed 0.50 cutoffs misalign precision/recall on uncalibrated logits.
- Must test on future dates (>= 2024-03-11) using validation-only thresholds.

4. Investigator Trust, XAI & Audit Parity

- SIU teams need local explanations & audit digests, not opaque scalars.
- Production deployment requires both server APIs & zero-server browser parity.

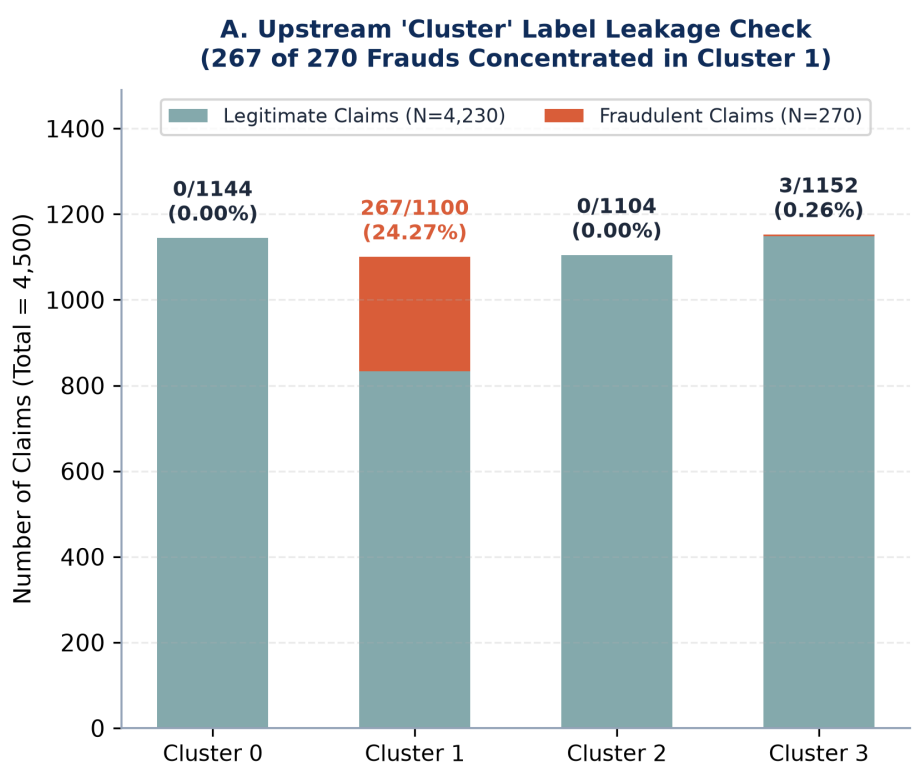
Literature Survey: Six Research Pillars & Identified Gaps

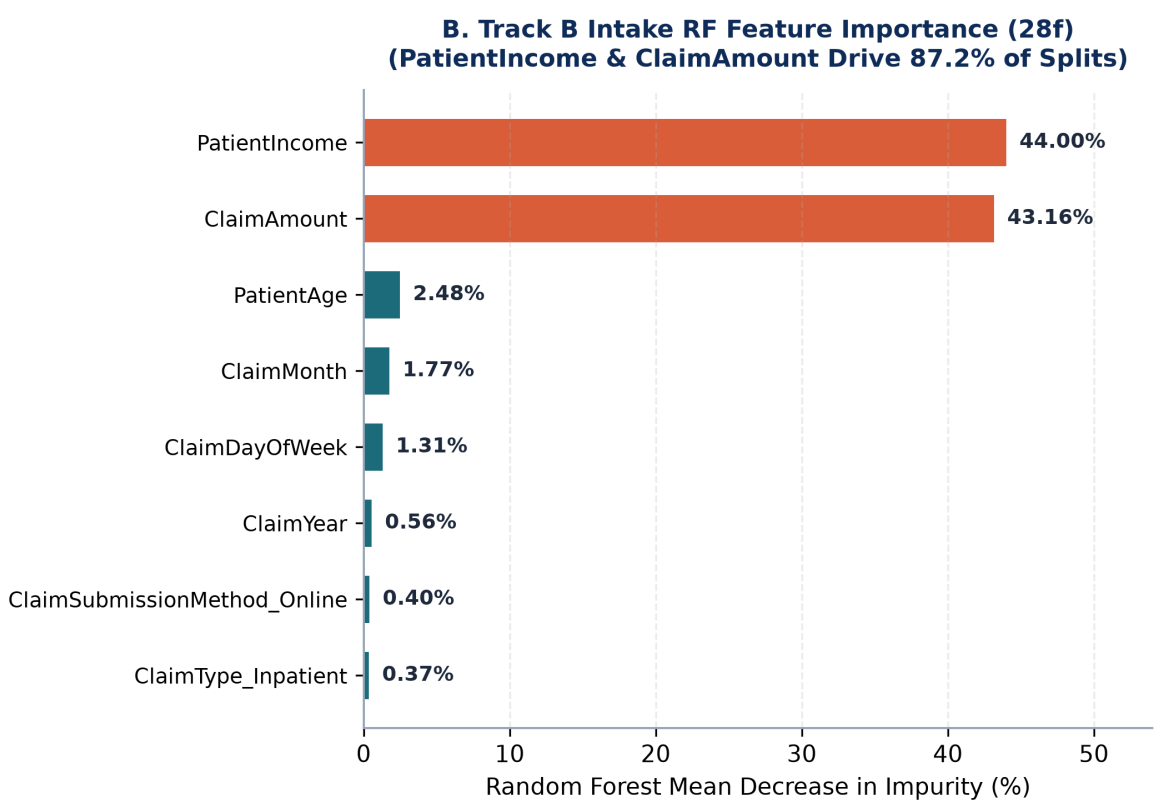
Synthesis Across 20 Peer-Reviewed & Foundational ML, DL, XAI, and Leakage Studies

Research Theme	Key Verified Citations	Core Insight in Literature	How Addressed in Our Project
Healthcare ML & Classical Ensembles	du Preez et al. (2025, 137 studies); Bauder & Khoshgoftaar (2017); Breiman (2001); Chen & Guestrin (2016)	Tree ensembles (RF, XGBoost, LightGBM) dominate tabular claims, but studies omit temporal checks.	Implemented 5 balanced classical models; audited random & chronological future splits.
Tabular Deep Learning vs. Trees	Johnson & Khoshgoftaar (2019); Huang et al. (2020); Gorishniy (2021); Grinsztajn (2022)	Attention improves neural tabular scoring, yet trees preserve sharp axis-aligned splits on ~10K rows.	Built MLP, ResidualMLP, CNN1D & TabTransformer; proved trees retain 1.00 PR-AUC on Intake.
Autoencoder Anomaly Detection	Sakurada & Yairi (2014); Chalapathy & Chawla (2019)	Reconstruction MSE fails when anomalies sit inside normal covariate convex support.	Trained normal-only PyTorch Autoencoder; openly documented AP=0.0523 negative result.
Metrics, Leakage & Shortcuts	Saito & Rehmsmeier (2015); Kaufman (2012); Geirhos (2020); Chawla (2002)	PR-AUC beats ROC-AUC under skew; leaky proxies and shortcuts invalidate offline scores.	Adopted PR-AUC + val-tuned F1; tested Cluster-free Intake & ClaimAmount-free No-Amount.
XAI, RAG & Agentic Compliance	Lundberg & Lee (2017, SHAP); Lewis et al. (2020, RAG); Kapadiya et al. (2022)	Human SIU review demands factor rationales, policy RAG grounding & tamper-evident hashes.	Built 6-agent UI, Gemini 2.0 Flash /api/chat co-pilot & XAI SHA-256 utility scaffold.

Dataset Audit: Structural Properties & Upstream Proxy Hazards

Health Insurance Fraud Claims.xlsx — 4,500 Claims x 19 Columns (Zero Missing, Zero Duplicates)





Critical Audit Takeaways Shaping Experimental Governance

- Upstream Cluster (0..3) places 267/270 frauds (98.89%) inside Cluster 1 -> Dropping Cluster tests realistic hospital intake (Track B Intake).
- ClaimAmount is >= $7,465.07 for every fraud (mean $8,733.57) vs. $100.12-$9,997.20 legit -> Dropping both tests amount dependency (No-Amount).

Mathematical Problem Formulation

Imbalance-Weighted Risk Minimization, Reconstruction MSE & Validation-Frozen Thresholding

Supervised Objective (Classical & DL)

• Inputs phi(r_i) in R^(d_s) built via train-only StandardScaler + OHE (29f / 28f / 27f).
• Weighted BCE: L = -(1/N_tr) sum [w+ y_i log sigma(z_i) + (1-y_i) log(1-sigma(z_i))], w+ = 15.67.
• AdamW optimization + validation PR-AUC early stopping.

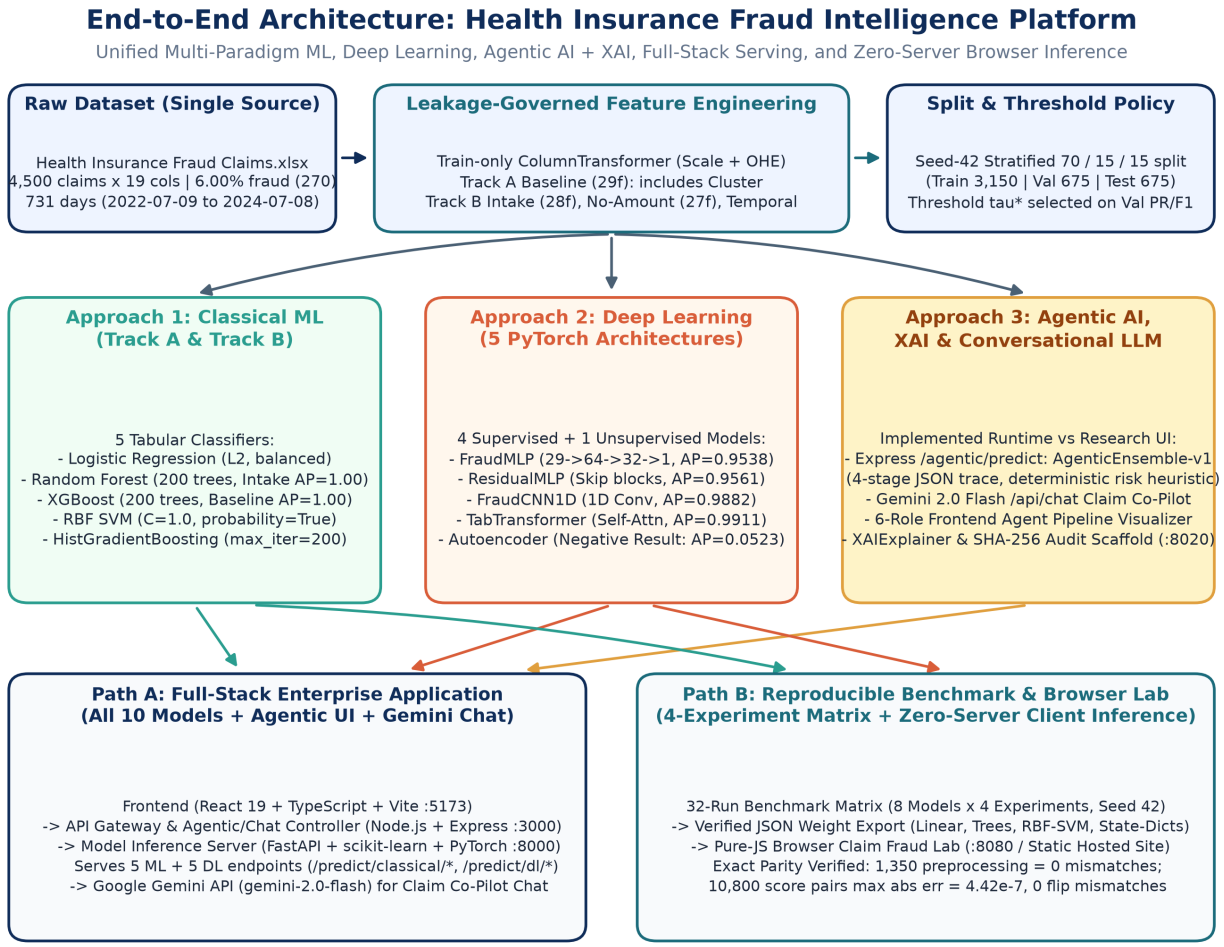
Unsupervised Autoencoder & Threshold Policy

• Autoencoder (29->16->8->16->29) fits strictly legitimate records (y=0) via MSE.
• Cutoff tau_AE = 0.128588 fixed at 95th percentile of validation errors.
• Track B cutoff: tau* = argmax_tau F1(y_val, I(p_val >= tau)), frozen on test.

Contract	Dim	Split Strategy	Train / Val / Test (Frauds)	Primary Hypothesis Tested
Baseline (Track A & B)	29	Stratified Seed 42 (70/15/15)	3,150 (189) / 675 (41) / 675 (40)	Unified 10-model comparison with original schema (includes Cluster).
Intake (Track B)	28	Stratified Seed 42 (70/15/15)	3,150 (189) / 675 (41) / 675 (40)	Realistic claim screening without upstream Cluster proxy.
No-Amount (Track B)	27	Stratified Seed 42 (70/15/15)	3,150 (189) / 675 (41) / 675 (40)	Stress test measuring non-monetary predictors when amount is withheld.
Temporal Intake	28	Chronological Date Split	3,152 (199) / 677 (34) / 671 (37)	Forward-in-time test (Train <= 2023-11-21; Test >= 2024-03-11).

Unified System Architecture: Approaches 1–3 & Dual Deployment

Coordinating Three Modeling Pillars Across Enterprise Microservices (Path A) and Browser Lab (Path B)



Team IIIT Dharwad (Supervisor: Prof. Ramesh Athe): B. Varshith (23BDS011, Approach 3 Agentic/Chat) | J. Ganesh Kumar Reddy (23BDS024, Approach 1 & Track B Lab) | M. Jagadeeswar (23BDS033, Approach 2 DL & Path A App)

Approach 1: Classical Machine Learning Pipelines

Five Diverse Classifiers Balancing Interpretability, Margin Separation & Nonlinear Tree Ensembles

Model	Inductive Bias & Key Hyperparameters	Track A PR-AUC	Track B Intake PR-AUC (F1)	Runtime Profile
LogisticRegression	L2 linear hyperplane; class_weight='balanced', max_iter=1000	91.86%	85.25% (0.7327)	~0.01s train; linear coefficients
RandomForest	200 bagged decision trees; class_weight='balanced', seed 42	99.94%	100.00% (0.9873)	0.61s train; Champion Intake Model
XGBoost	200 boosted trees; depth 5, lr 0.05, scale_pos_weight=15.67	100.00%	100.00% (0.9877)	0.17s train; 100% Test Recall
SVM (RBF Kernel)	Radial basis margin + Platt scaling; class_weight='balanced'	92.52%	81.81% (0.7711)	Drops -10.7 AP without Cluster
HistGradientBoosting	Histogram-binned boosting; max_iter=200, lr=0.08	97.65%	97.65% (0.9873)	1.0000 PR-AUC & F1 on Temporal

Why Orthogonal Tree Ensembles Dominate Structured Insurance Claims

- Capture sharp boundary thresholds naturally (ClaimAmount interacting with PatientIncome & ProviderSpecialty) without feature scaling distortion.
- Maintain robustness across 22 sparse one-hot categorical indicators where smooth kernels (RBF SVM) & hyperplanes degrade once Cluster is dropped.

Approach 2: Five PyTorch Deep Learning Architectures

Comparing Feedforward, Residual, 1D-Convolutional, Self-Attention & Unsupervised Reconstruction Models

1. FraudMLP & 2. ResidualMLP

• FraudMLP: Linear(d->64)->BN->ReLU->Drop(0.2)->Linear(64->32)->BN->ReLU->1 (0.73ms).
• ResidualMLP: Pre-activated identity skip blocks improving gradient flow (0.67ms, AP=95.61%).

3. FraudCNN1D (Highest DL F1)

• Reshapes aligned 29f/28f vectors into 1D channel sequence for local convolution filters.
• Captures adjacent numeric/OHE interactions; attains 98.82% PR-AUC & 90.91% F1 (0.77ms).

4. TabTransformer (Best DL PR-AUC)

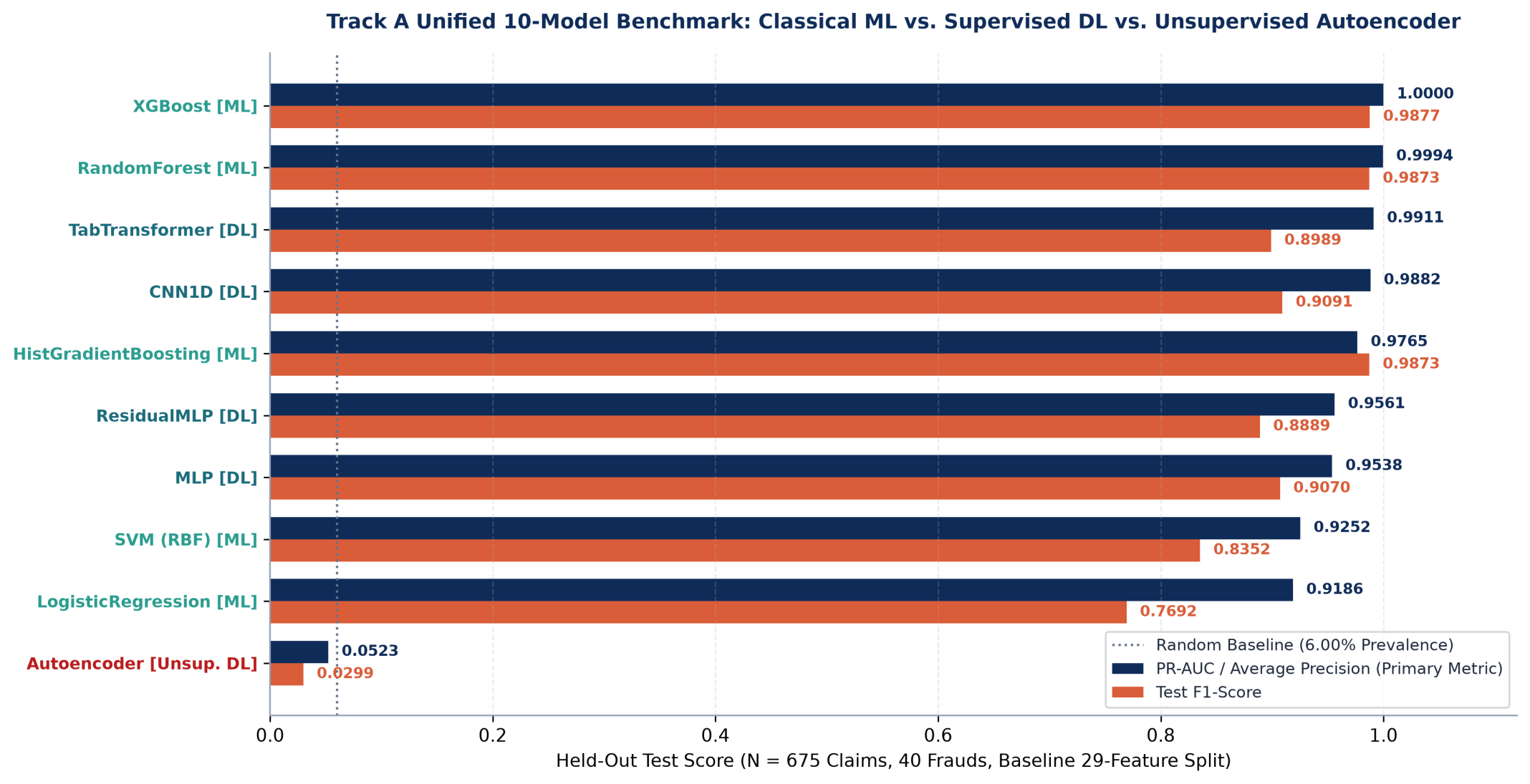
• Projects discrete categoricals via Multi-Head Self-Attention contextual embeddings.
• Leads all DL architectures at 99.11% PR-AUC & 99.94% ROC-AUC (18.65ms batch latency).

5. Unsupervised Autoencoder (29 -> 16 -> 8 -> 16 -> 29): Controlled Negative Result

• Trained exclusively on legitimate training records (y = 0) minimizing Mean Squared Reconstruction Error without fraud labels.
• Validation 95th percentile cutoff tau_AE = 0.128588 yielded PR-AUC = 5.23% (below 6.0% base rate), ROC-AUC = 46.02%, F1 = 2.99% (TP=1, FN=39, FP=26).
• Scientific Root Cause: Frauds lie inside legitimate covariate support (ClaimAmount 7,465-9,978 sits within legit 100-9,997), so normal encoder-decoder reconstructs them accurately.

Results — Track A Unified 10-Model Test Benchmark (29f)

Held-Out Test Split (N = 675 Claims, 40 Frauds, Standard Cutoff 0.50 / AE MSE 0.1286)



Approach 3: Agentic Triage, XAI & Gemini Co-Pilot

Honest Architecture Separating Verified Runtime Endpoints from Research UI & Future LangGraph/RAG

Approach 3: Agentic AI, Explainability (XAI), and Conversational Co-Pilot Status

Transparent Separation of Implemented Runtime Routes, Research/UI Scaffolds, and Production Roadmap

IMPLEMENTED & VERIFIED RUNTIME PATHS (Node.js Express :3000 + React Vite :5173)

1. ClaimInputForm & AgenticPage.tsx

9 UI fields submitted via
POST /agentic/predict
+ /api/chat Co-Pilot Drawer

2. AgenticEnsemble-v1 (src/agentic/predict.ts)

4 Trace Events emitted:
ingestion -> validation ->
risk-scorer -> xai-explainer
(Deterministic heuristic formula)

3. Gemini Co-Pilot (src/routes/chat.ts)

POST /api/chat invokes
Google gemini-2.0-flash
via GEMINI_API_KEY
for claim Q&A & triage

4. Direct ML/DL UI (ClassicalML & DL Pages)

Live FastAPI :8000 inference
for all 10 trained models
+ browser offline fallback

RESEARCH SCAFFOLD & 6-AGENT VISUAL DESIGN (Approach-3-Agentic-AI :8020 + agentic/data/agents.ts)

1. Data Collector

Schema checks
(UI Stage Card)

2. Claim Analyzer

Amount & codes
(UI Stage Card)

3. Pattern Detector

Rule anomaly
(UI Stage Card)

4. Billing Review

Policy fit view
(UI Stage Card)

5. Provider Profiler

Outlier profile
(UI Stage Card)

6. Coordinator + XAI

XAIExplainer
SHA-256 + SHAP
approx scaffold

DOCUMENTED ENGINEERING GAPS & NEXT-PHASE PRODUCTION ROADMAP (Planned vs Implemented Transparency)

Live ML-Agent Binding

Replace heuristic score in /agentic/predict
with real FastAPI (:8000) calls to
Intake RandomForest / XGBoost / TabTransformer

OCR / VLM & RAG Grounding

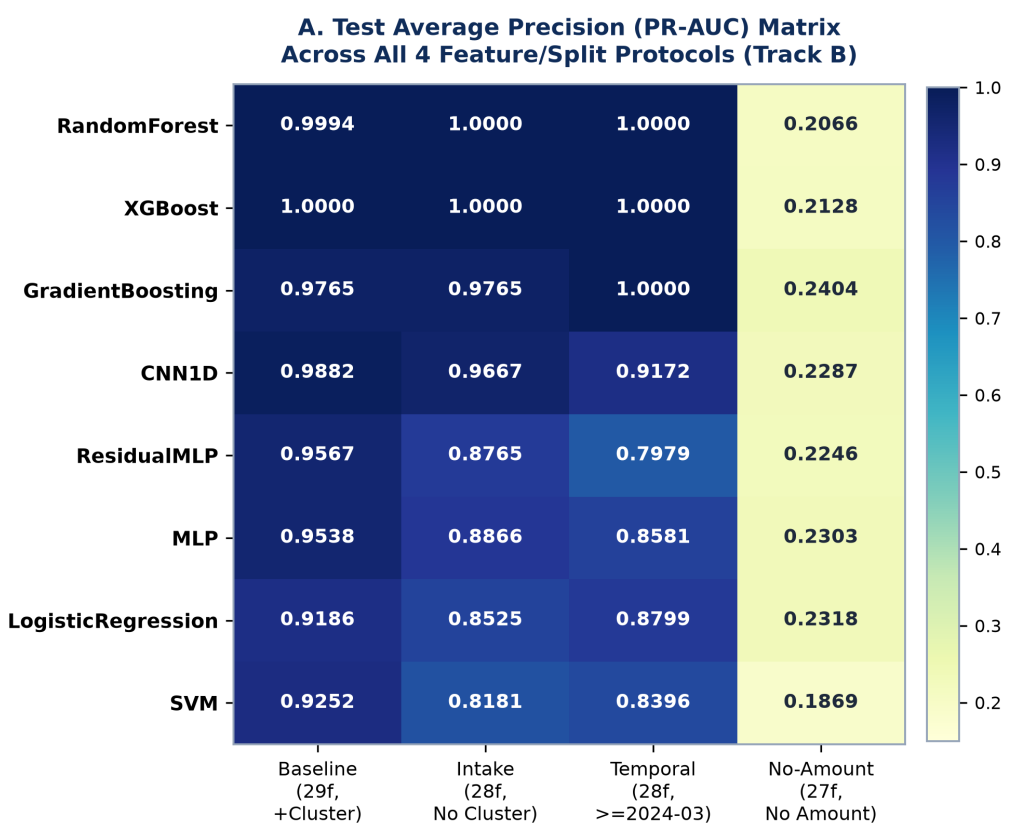
Connect medical bill PDF/image OCR
and ChromaDB/FAISS clinical policy RAG
for evidence-cited SIU audit packets

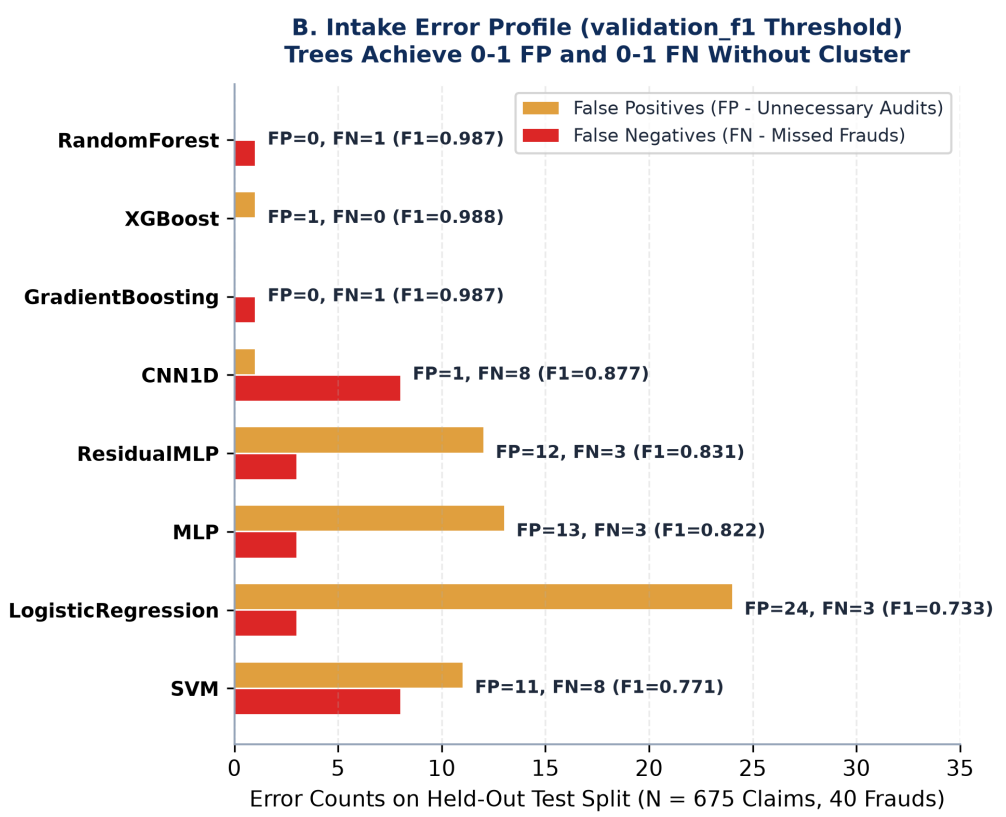
Persistent Audit DB & Calib.

Persist SHA-256 audit hashes in PostgreSQL
and calibrate classifier probabilities
for expected financial loss optimization

Results — Track B Sensitivity & Chronological Ablations

32 Runs (8 Models x 4 Protocols) Quantifying Cluster Leakage, Temporal Stability & Amount Dependence





Key Findings from 32 Controlled Runs

• Trees ignore Cluster removal (RF/XGB 1.00 PR-AUC) while SVM/MLPs drop 6.6-10.7 AP; Trees reach 1.00 F1 chronologically; No-Amount collapses all models to 0.19-0.24 AP.

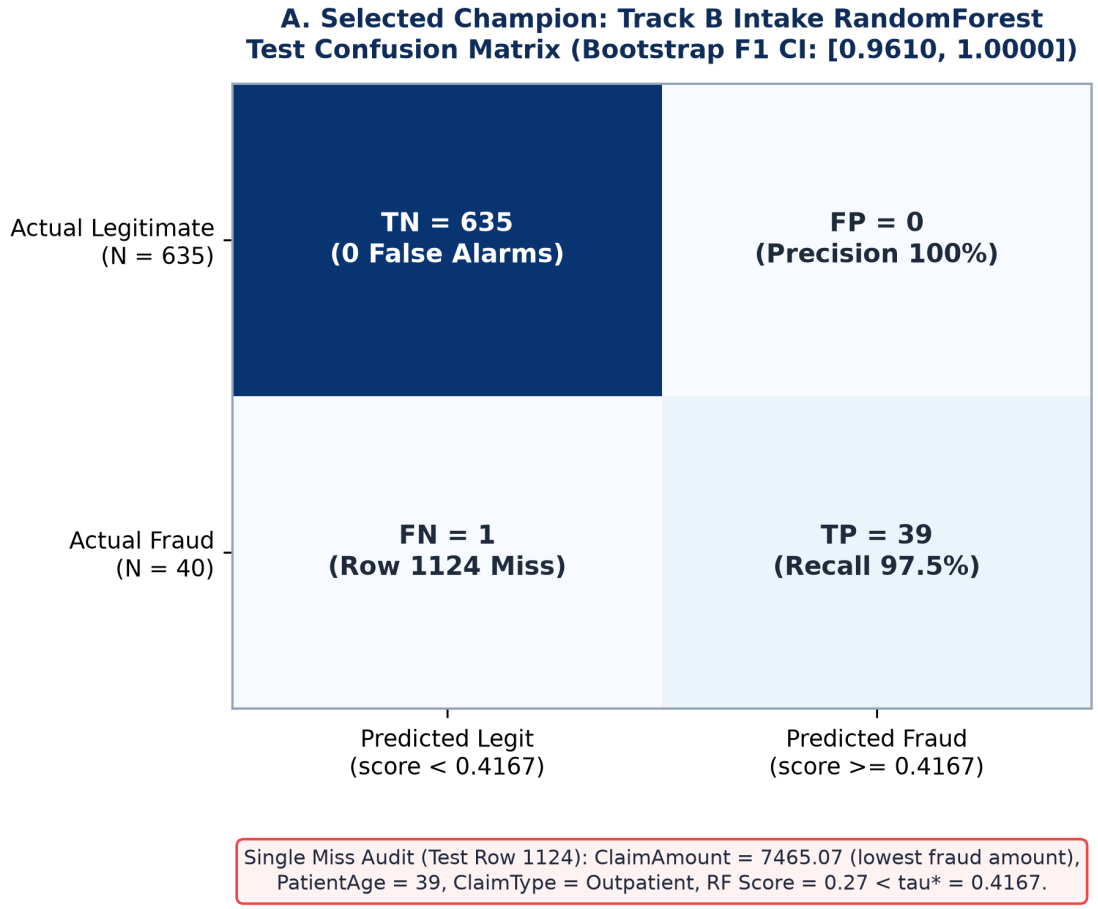
Track B Quantitative Ledger Across All 4 Experimental Protocols

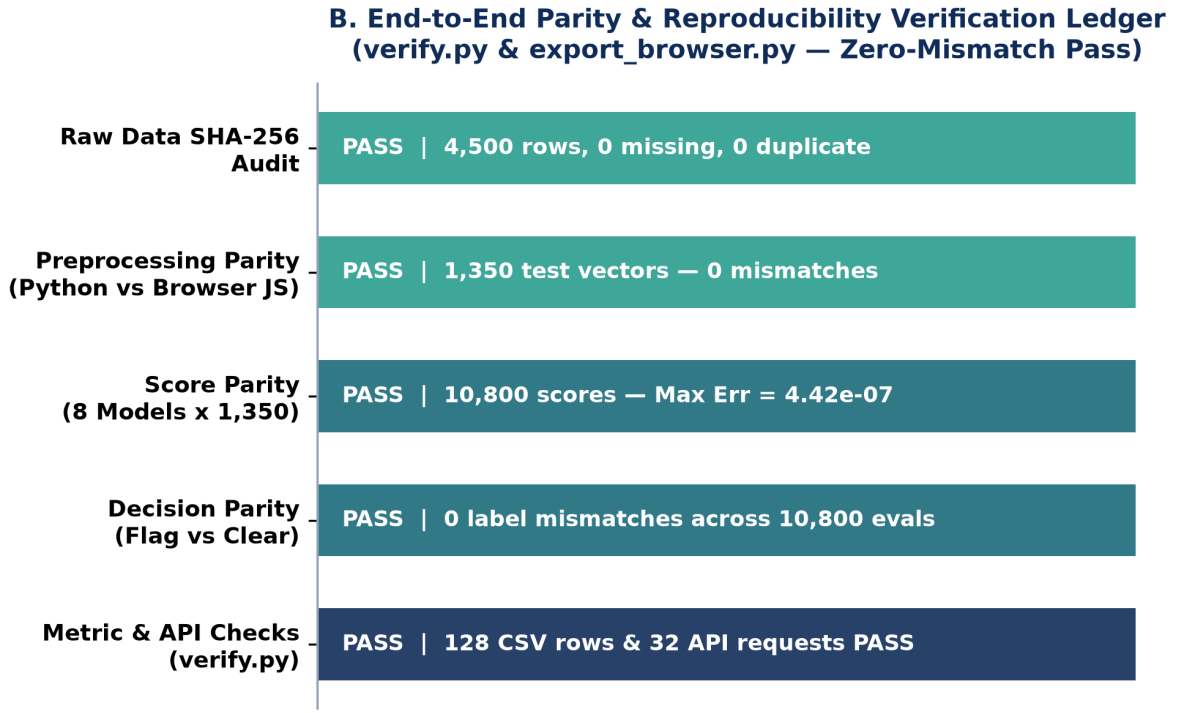
Test PR-AUC (Average Precision) and Validation-F1 Tuned F1 Score on Held-Out Splits

Model	Baseline (29f) AP / F1	Intake (28f) AP / F1	No-Amount (27f) AP / F1	Temporal (>=2024-03) AP / F1	Intake Threshold & Errors
RandomForest (Default)	0.9994 / 0.9750	1.0000 / 0.9873	0.2066 / 0.3636	1.0000 / 1.0000	tau*=0.4167 | 0 FP, 1 FN
XGBoost	1.0000 / 0.9877	1.0000 / 0.9877	0.2128 / 0.3524	1.0000 / 1.0000	tau*=0.2121 | 1 FP, 0 FN
GradientBoosting (HGB)	0.9765 / 0.9873	0.9765 / 0.9873	0.2404 / 0.3636	1.0000 / 1.0000	tau*=1.0000 | 0 FP, 1 FN
CNN1D (Best Track B DL)	0.9882 / 0.9268	0.9667 / 0.8767	0.2287 / 0.2788	0.9172 / 0.8108	tau*=0.9691 | 1 FP, 8 FN
ResidualMLP	0.9567 / 0.9286	0.8765 / 0.8315	0.2246 / 0.3438	0.7979 / 0.7229	tau*=0.2751 | 12 FP, 3 FN
MLP	0.9538 / 0.9136	0.8866 / 0.8222	0.2303 / 0.3077	0.8581 / 0.6957	tau*=0.8359 | 13 FP, 3 FN
LogisticRegression	0.9186 / 0.7835	0.8525 / 0.7327	0.2318 / 0.3191	0.8799 / 0.7397	tau*=0.7618 | 24 FP, 3 FN
SVM (RBF)	0.9252 / 0.8974	0.8181 / 0.7711	0.1869 / 0.3041	0.8396 / 0.7619	tau*=0.5859 | 11 FP, 8 FN

Champion Forensics, Bootstrap CIs & Browser Parity Verification

Inspecting Row 1124 & Verifying 10,800 Score Comparisons Within 4.42e-7 Tolerance





Path A Implementation: Three-Tier Enterprise Web Platform

React 19 UI (:5173) -> Node.js/Express Gateway (:3000) -> FastAPI ML/DL Server (:8000)

Tier 1: React 19 + TypeScript UI (:5173)

• Pages: HomePage overview, ClassicalMLPage (5 models + offline fallback), DLPage (all 5 DL models + gauge), AgenticPage (6-role cards).
• Interactive Gemini Co-Pilot drawer (ChatBox.tsx) explaining risk outputs in plain English.
• Submits unified claim JSON schema across all UI tabs.

Tier 2: Node.js + Express (:3000)

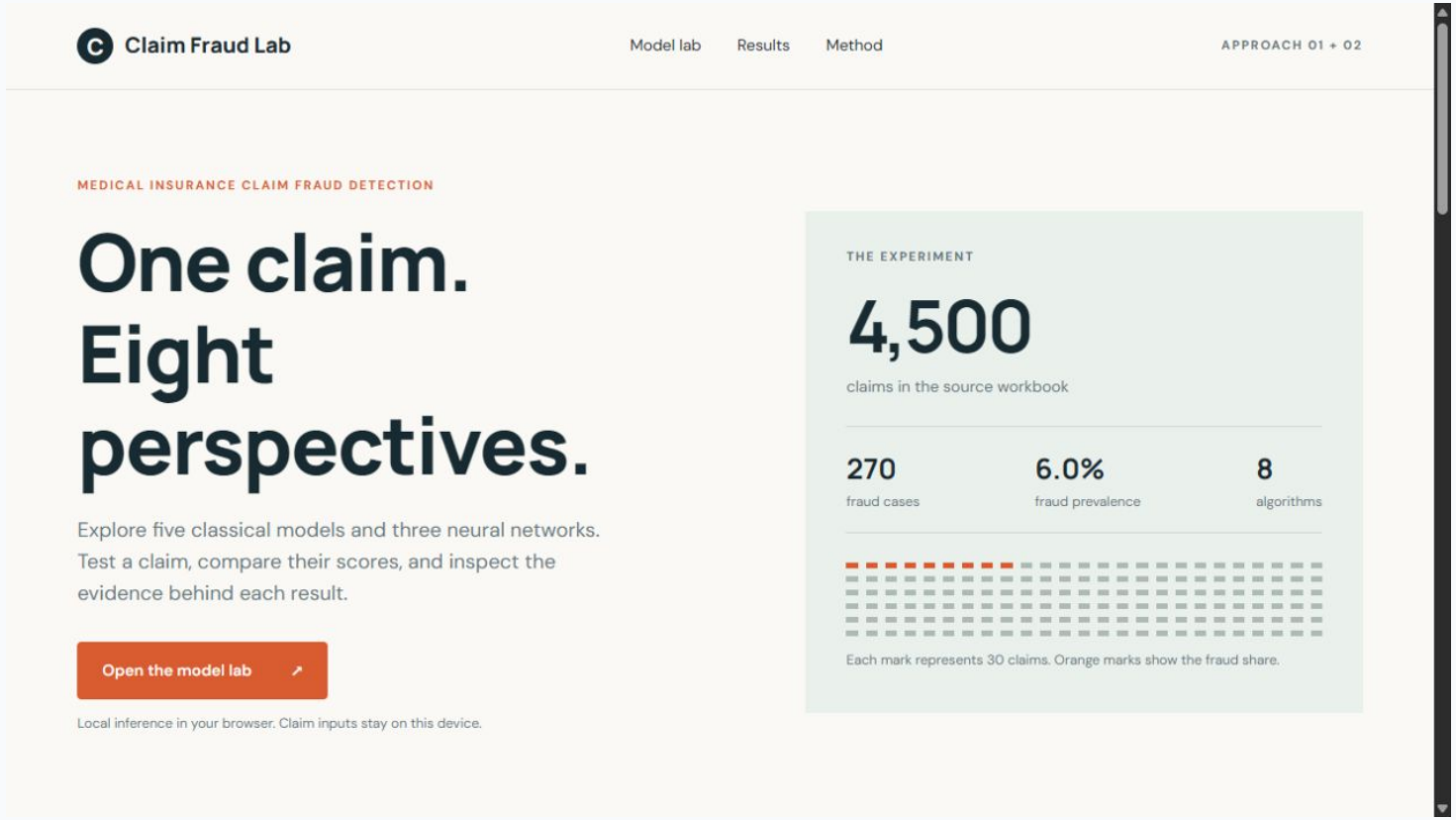
• Routes /api/ml/* proxy cleanly to FastAPI :8000 classical & DL prediction endpoints.
• Active /agentic/predict serves deterministic 4-stage AgenticEnsemble-v1 audit trace.
• Route /api/chat invokes Google @google/genai (gemini-2.0-flash) with claim context.

Tier 3: Python FastAPI Server (:8000)

• classical_service.py loads sklearn/XGBoost pipelines via joblib across 5 endpoints.
• dl_service.py loads frozen 29f scaler & PyTorch .pt checkpoints for MLP, ResidualMLP, CNN1D, TabTransformer & Autoencoder.
• Pydantic validation enforces strict types & bounds.

Path B Innovation: Zero-Server Browser Claim Fraud Lab (:8080)

Exact Pure-JavaScript Execution of All 8 Supervised Models Without Backend Round-Trips



Export & Parity Guarantees

- export_browser.py compiles StandardScaler means/scales, OHE categories, Logistic weights, RF/XGB/HGB decision trees, RBF support vectors & neural layers into JSON.
- Runs live locally on :8080 or as a zero-backend static web bundle.
- Verified via verify.py across 1,350 preprocessing rows (0 errors) & 10,800 score pairs (max abs diff 4.42e-7, 0 label flips).

Summary of Key Project Innovations

Five Distinct Engineering & Scientific Advances Beyond Baseline Notebooks

1. Leakage-First Feature Governance

• Dual-track architecture isolates Cluster & ClaimAmount shortcuts via explicit 4-protocol matrix.

2. Honest Heterogeneous 10-Model Evaluation

• Includes both TabTransformer self-attention (99.11% AP) and a transparent Autoencoder negative result (5.23% AP).

3. Exact Python-to-Browser Parity Compiler

• Zero-server JS implementation matches PyTorch, XGBoost, RF & RBF SVM within 4.42e-7.

4. Validation-Only Thresholds & Bootstrap CIs

• No test peeking: thresholds optimize F1 strictly on val folds, paired with 1,000-rep CIs.

5. Modular Three-Tier Stack + Gemini 2.0 Flash Co-Pilot

• Combines FastAPI (:8000) checkpoints, Express (:3000) Agentic/Chat gateway, React (:5173) UI & SHA-256 XAI hashing.

Complete Deliverables & Documentation Suite (PDF + LaTeX + Office)

All Reports, IEEE Manuscript, Code Modules & Documentation Unified

Deliverable Category	Generated Workspace Artifact(s)	Contents & Format Fidelity
Literature Survey	docs/01_LITERATURE_SURVEY.md + deliverables/01_Literature_Survey.pdf	6 thematic research pillars, 20 verified DOIs, synthesis gap matrix.
Problem Formulation	docs/02_PROBLEM_FORMULATION.md + deliverables/02_Problem_Formulation.pdf	Dataset audit, weighted BCE, Autoencoder MSE, threshold & parity contracts.
Innovation & Implementation	docs/03_INNOVATION_AND_IMPLEMENTATION.md + deliverables/03_Innovation_and_Implementation.pdf	5 core innovations + Approach 1-3 blueprints + Path A/B deployment.
Full Documentation	docs/04_FULL_PROJECT_DOCUMENTATION.md + deliverables/04_Full_Project_Documentation.pdf	Module map, Track A & B metrics, runbook, REST API spec, limitation log.
IIIT Dharwad Report	deliverables/MiniProjectReport_LaTeX/ + deliverables/MiniProjectReport.pdf	Built from MiniProjectTemplate.zip (Title, Certificate, ToC/LoF/LoT, Ch 1-5, BibTeX).
IEEE Conference Paper	deliverables/IEEE_Paper_LaTeX/ + IEEE_Paper.pdf + IEEE_Paper.docx	IEEEtran two-column manuscript in LaTeX, compiled PDF & editable DOCX.
Presentation & Notes	deliverables/Project_Presentation.pptx + Project_Presentation.pdf + Notes.md	20-slide widescreen deck in editable PPTX & inspected PDF + speaker script.

Audited Limitations & Next-Phase Production Roadmap

Engineering Transparency Balancing Current Strengths with Concrete Future Upgrades

Observed Boundaries & Honest Limitations

• Synthetic Corpus Reliance on ClaimAmount: Removing ClaimAmount collapses AP from 1.00 to 0.19-0.24 because zero frauds exist below $7,465.07.
• Uncalibrated Classifier Scores: Neural pos_weight=15.67 and raw RF proportions require CalibratedClassifierCV for expected-loss calculation.
• Approach 3 Runtime Gap: Express /agentic/predict uses deterministic heuristic rather than live calls to :8000 ML/DL endpoints.
• Document & Database Gap: Clinical bill OCR/VLM parser, policy RAG & PostgreSQL remain planned phase-2 extensions.

Next-Phase Engineering Roadmap

• 1. Bind /agentic/predict to FastAPI (:8000) Intake RandomForest + TabTransformer scores with SHAP explanations.
• 2. Calibrate probabilities (isotonic/Platt scaling) on held-out validation splits for financial triage optimization.
• 3. Integrate medical invoice OCR/VLM + ChromaDB clinical policy RAG for evidence-grounded SIU packets.
• 4. Store immutable SHA-256 audit hashes in PostgreSQL & evaluate on multi-institutional clinical corpora.

Conclusion & Key Takeaways

• Data discipline matters as much as architecture: isolating Cluster leakage exposes non-tree fragility immediately.

• Random Forest & XGBoost lead realistic Intake & Chronological rollouts (1.0000 PR-AUC; 0.9873 Random / 1.0000 Temporal F1).

• TabTransformer (0.9911 PR-AUC) & CNN1D (0.9882 PR-AUC) lead supervised DL; Autoencoder negative result (0.0523 AP) proves manifold overlap.

• Both an enterprise 3-tier app (React + Node + FastAPI + Gemini Chat) and a verified 4.42e-7-parity browser lab are operational.

Thank You — Questions & Live System Demonstration